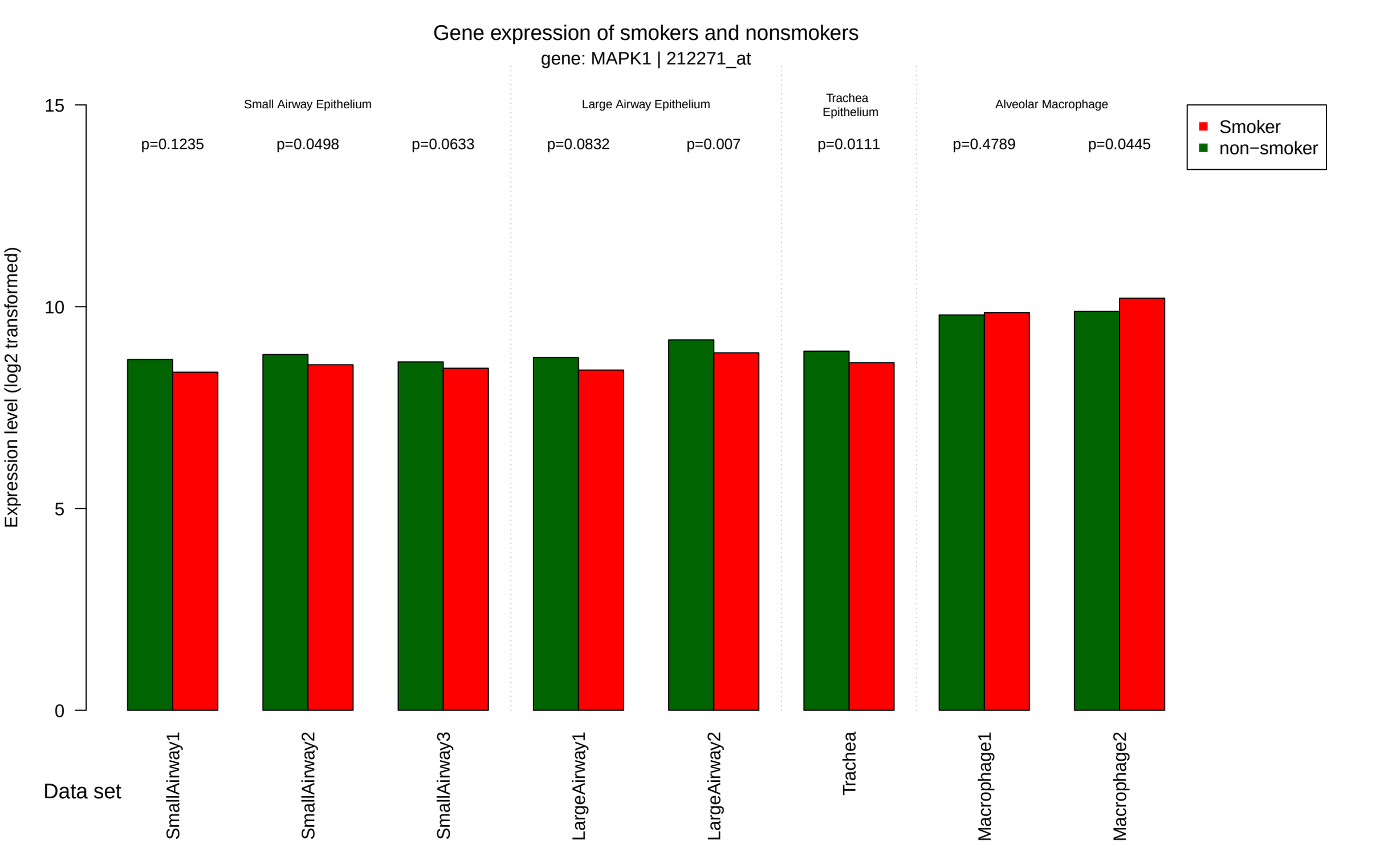
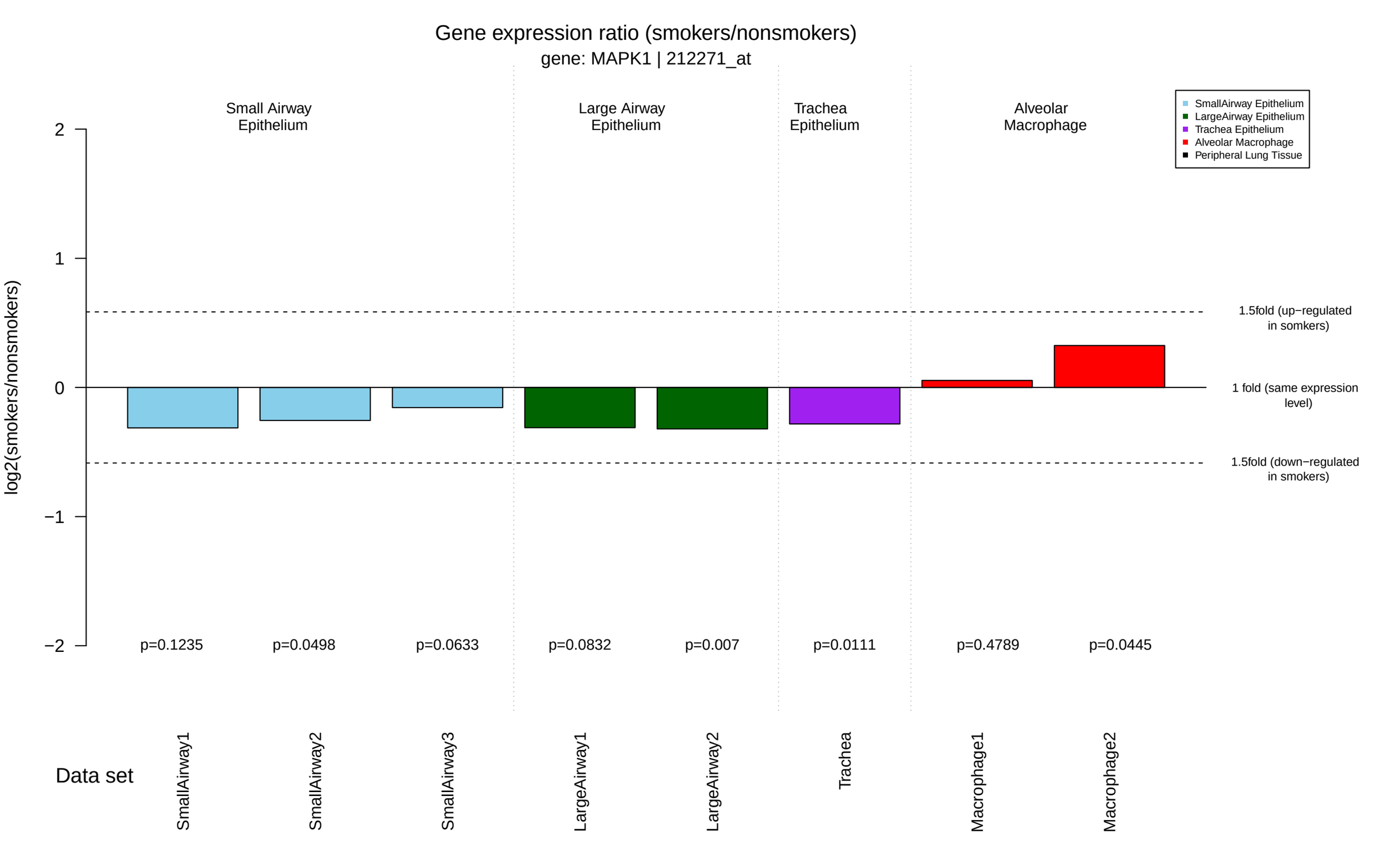
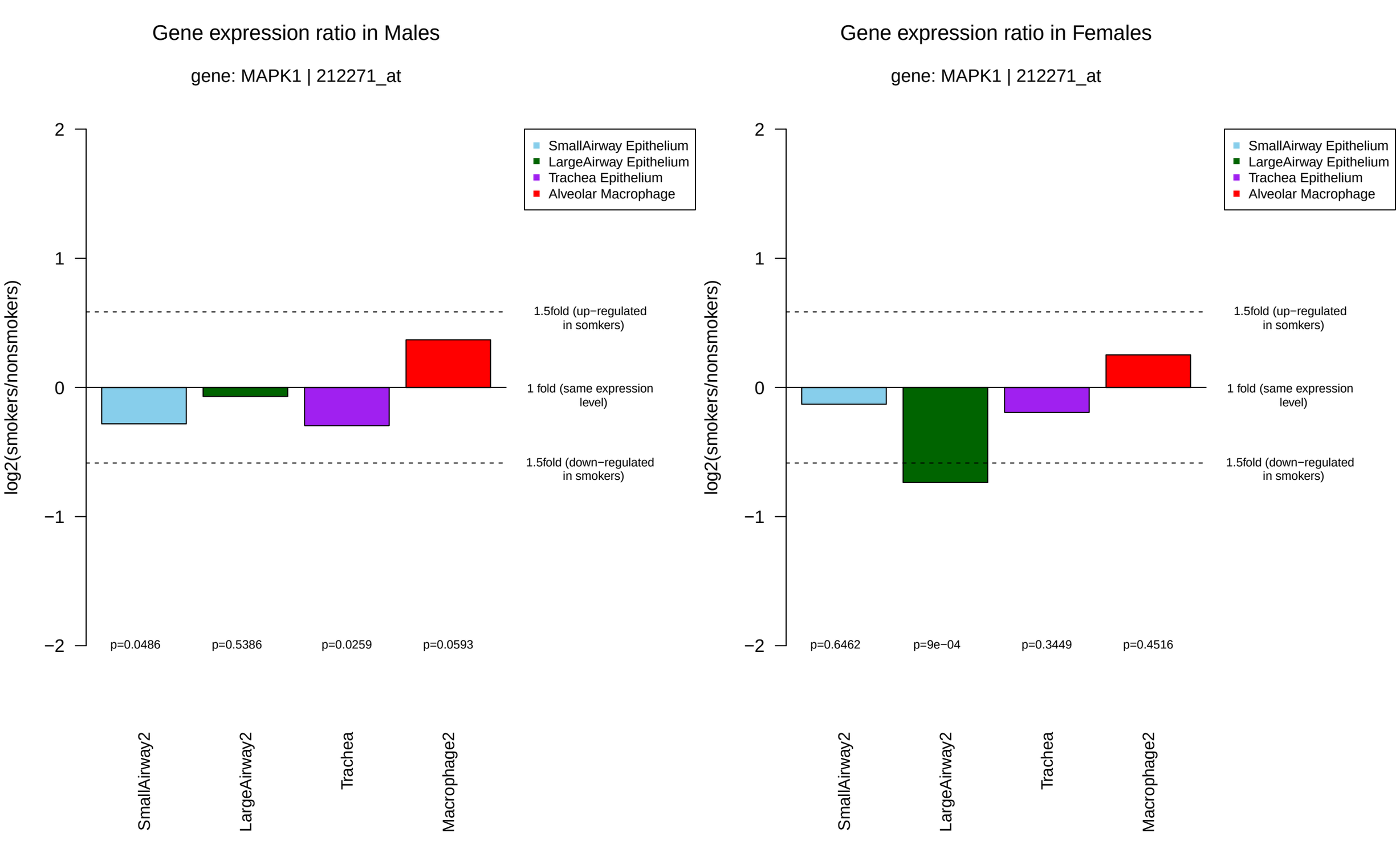
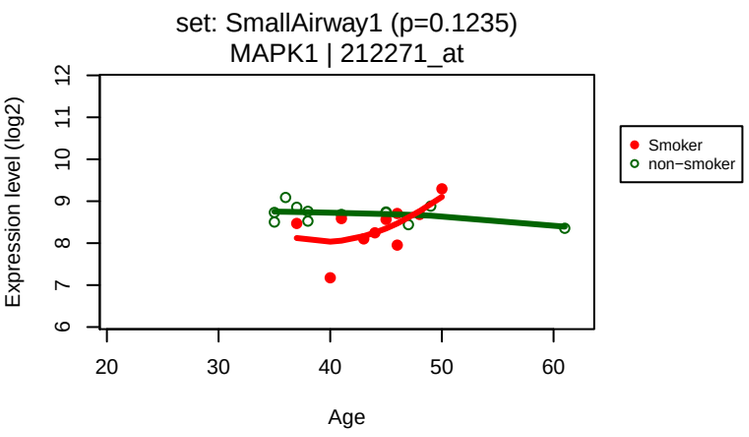
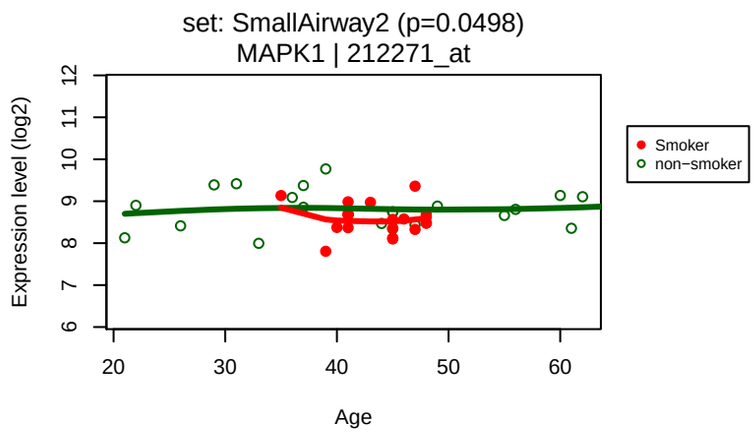
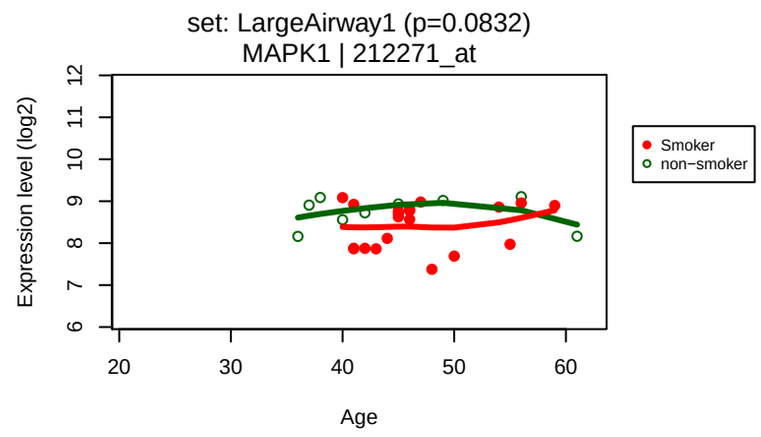

Gene expression of smokers and nonsmokers
gene: MAPK1 | 212271_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1235
p=0.0498
p=0.0633
p=0.0832
p=0.007
p=0.0111
p=0.4789
p=0.0445
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

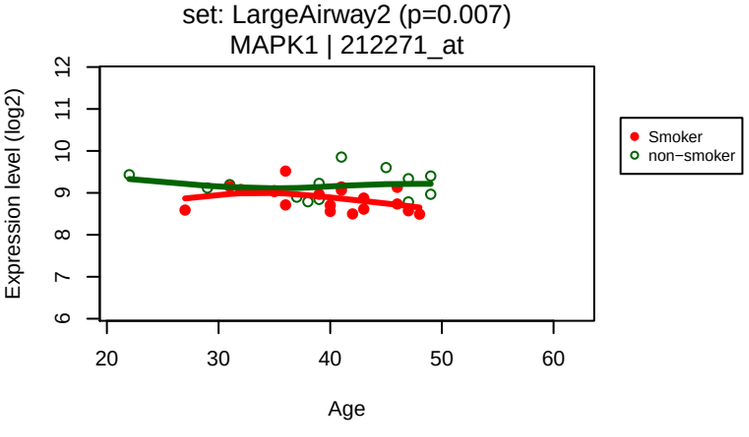

Gene expression ratio (smokers/nonsmokers)
gene: MAPK1 | 212271_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1235
p=0.0498
p=0.0633
p=0.0832
p=0.007
p=0.0111
p=0.4789
p=0.0445
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

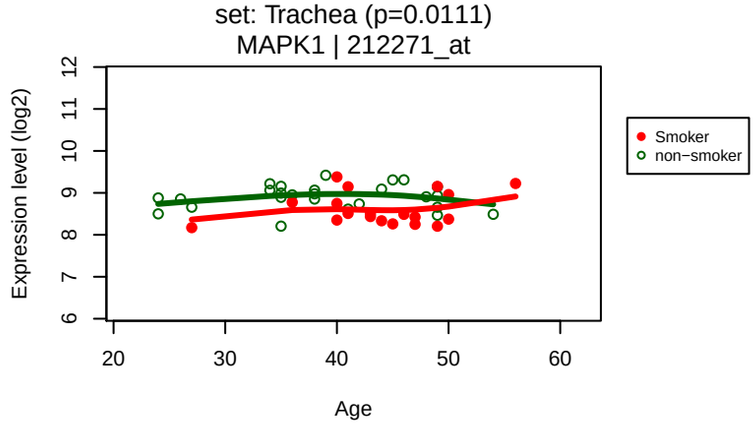

Gene expression ratio in Males
gene: MAPK1 | 212271_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0486
p=0.5386
p=0.0259
p=0.0593
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPK1 | 212271_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6462
p=9e−04
p=0.3449
p=0.4516
SmallAirway2
LargeAirway2
Trachea
Macrophage2

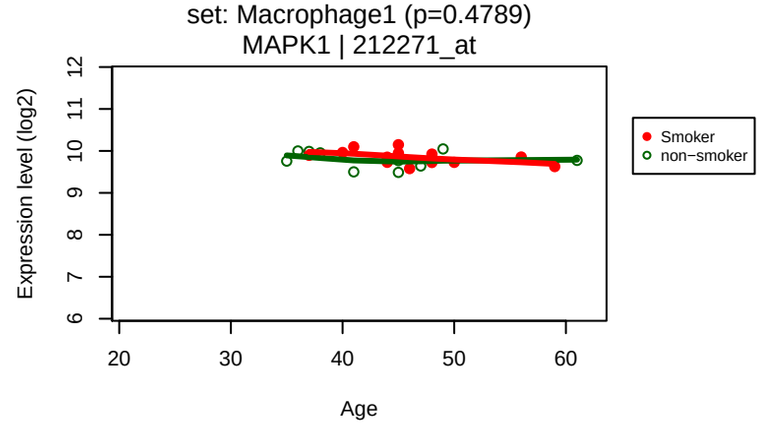

set: SmallAirway1 (p=0.1235)
MAPK1 | 212271_at
Expression level (log2)
Age
Smoker
non−smoker

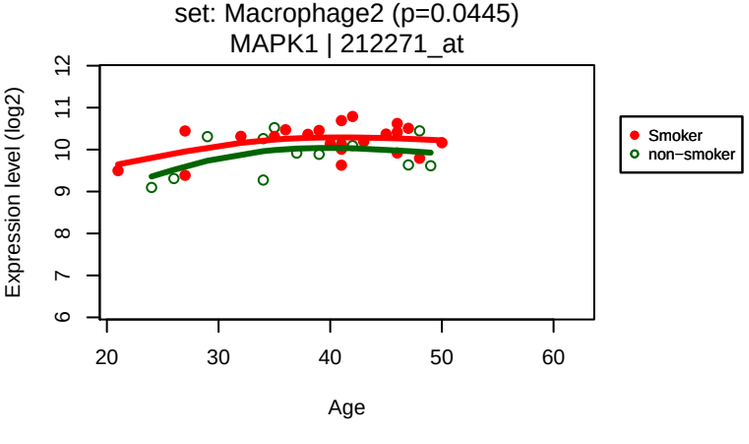

set: SmallAirway2 (p=0.0498)
MAPK1 | 212271_at
Expression level (log2)
Age
Smoker
non−smoker

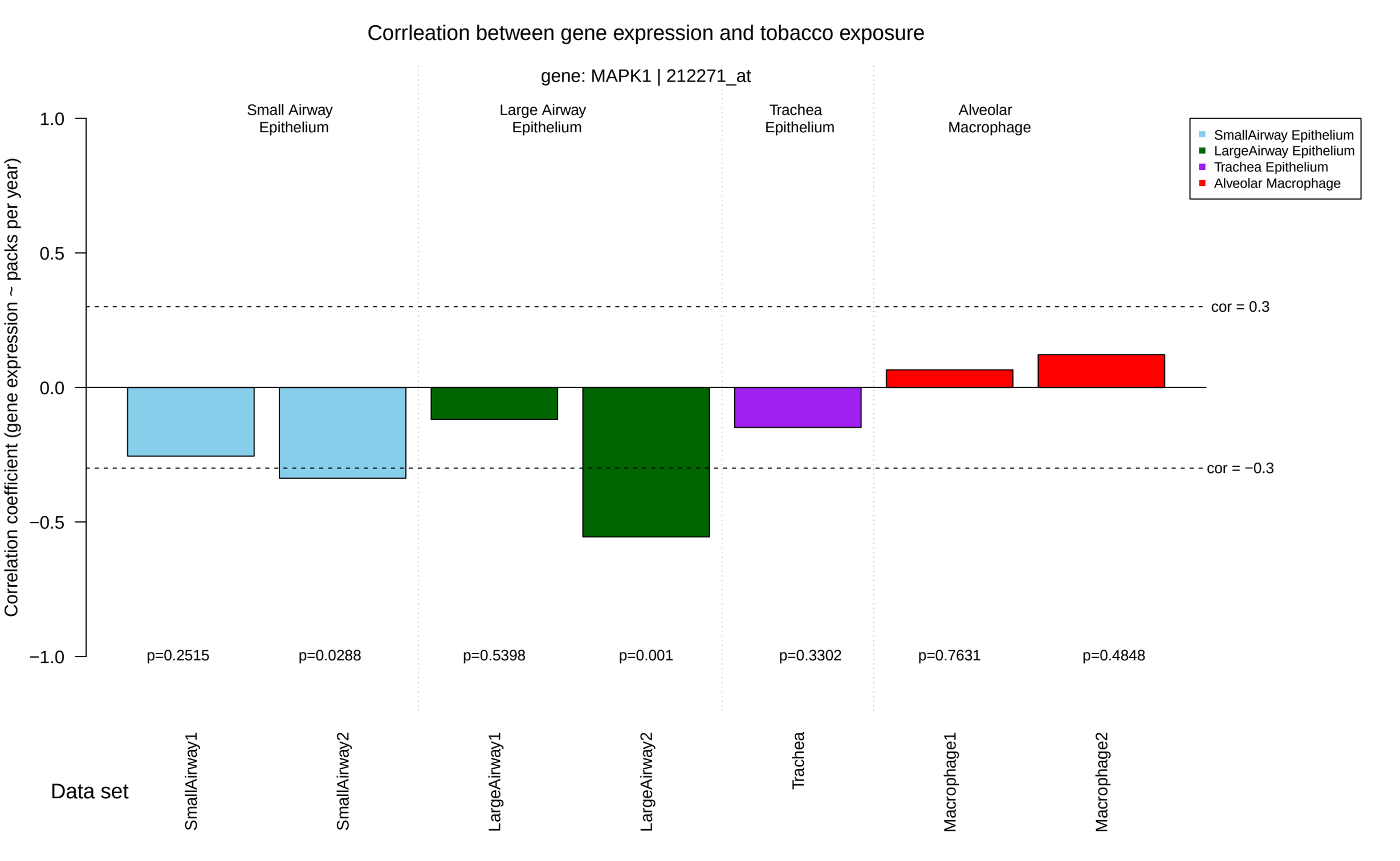

set: LargeAirway1 (p=0.0832)
MAPK1 | 212271_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.007)
MAPK1 | 212271_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0111)
MAPK1 | 212271_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4789)
MAPK1 | 212271_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0445)
MAPK1 | 212271_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPK1 | 212271_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2515
p=0.0288
p=0.5398
p=0.001
p=0.3302
p=0.7631
p=0.4848
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2